MATH 2101 LINEAR ALGEBRA I
FALL SEMESTER 2025
TEST 1

Last Name:

First Name:

UID:

Instructons:

(1) The question paper is in the **last page**. You may tear off the last page, but **return the questions paper** at the end of the test.
(2) Write answers in the attached A4 papers. Writings in draft papers will NOT be graded.
(3) Show your steps or reasonings clearly to receive full credits unless otherwise specified in the question.

Write your work in this page

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Question Paper (Fall 2025)

Name:
UID:

(1) (4 points) Find the inverse of the following matrix using *adjugate matrix*:

$$\begin{pmatrix} 1 & 2 & 0 \\ -1 & 0 & 1 \\ 0 & 2 & -1 \end{pmatrix}.$$

(2) (6 points) Let V be a vector space. Determine whether the following statements are true or false. Briefly explain your answer.

(a) If S is a linearly dependent subset in V, then each vector in S is a linear combination of other vectors in S.

(b) Subsets of linearly dependent sets in V are also linearly dependent.

(3) (4 points) Let V be the vector space of functions from $\mathbb{R}$ to $\mathbb{R}$. Define

$$S = \{f : \mathbb{R} \to \mathbb{R} | f(1) = 1\}.$$

Determine whether S forms a subspace of V. Explain your answer.

(4) (5 points) Given the following system of linear equations

$$2x_1 + 3x_2 + ax_3 = 1$$

$$4x_1 + 6x_2 + bx_3 = 2$$

What can one say about the number of solutions for the system for different a and b? Explain your answer.

(5) (6 points) Let A be an $n \times m$ matrix.

(a) Suppose $n = m$ and A is invertible. Find the reduced row echelon form of

$$\begin{pmatrix} A & B \\ 0 & A \end{pmatrix}$$

for some $n \times m$ matrice B. Explain your answer.

(b) Let r be the number of leading ones in the RREF of A. What is the maximum possible number of leading ones in the reduced row echelon form of

$$\begin{pmatrix} A & B \\ 0 & A \end{pmatrix}.$$